

SEEING THE PATTERNS, CLOSING THE LOOP: EVOLVING ENVIRONMENTS FOR AGENT TRAINING

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ABSTRACT

LLM agents can serve as data researchers, automating the creation of training data for LLM post-training. We study data-centric recursive self-improvement for agent post-training. In this setting, a researcher agent starts from an empty environment library and constructs environments for online RL without access to the target backend or verifier; evaluation of the updated model then supplies feedback for the next round. Our experiments and audits identify three challenges: incomplete behavioral analysis, disconnected data–training–evaluation feedback, and incomplete implementation of intended environment mechanisms. We introduce ENVOPT, which uses source-traceable hierarchical analysis to *see the patterns* in agent behavior and target-conditioned context refinement to *close the loop* between data, training, and evaluation. These two designs guide environment construction and revision, supported by specification-driven implementation and independent review. Preliminary experiments with an earlier implementation yield observed test gains of 16.67 and 11.11 percentage points on BFCL and ACEBench, respectively, although gains vary across rounds. These findings suggest that evidence-guided environment evolution is a promising direction for automated agent post-training.

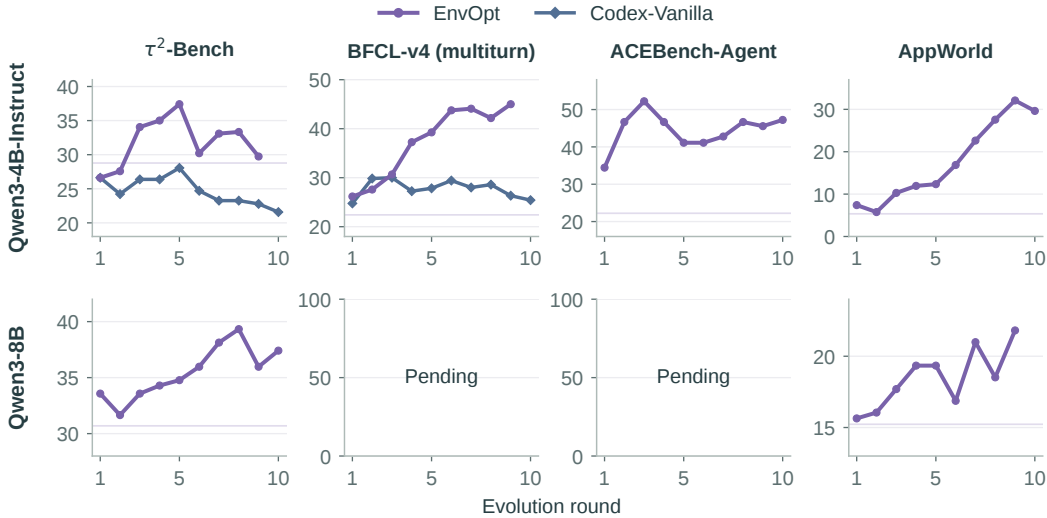


Figure 1: Public-evaluation performance over iterations 1–10, comparing ENVOPT with available Codex-Vanilla results; unobserved iterations are left blank. Qwen3-8B curves show completed evaluations for τ^2 and AppWorld (TGC), while its BFCL and ACEBench results are pending.

1 INTRODUCTION

Recursive self-improvement (RSI) asks whether AI systems can use evidence from one development cycle to improve the next. MetaRSI[CITE: MetaRSI] distinguishes three writable surfaces for im-

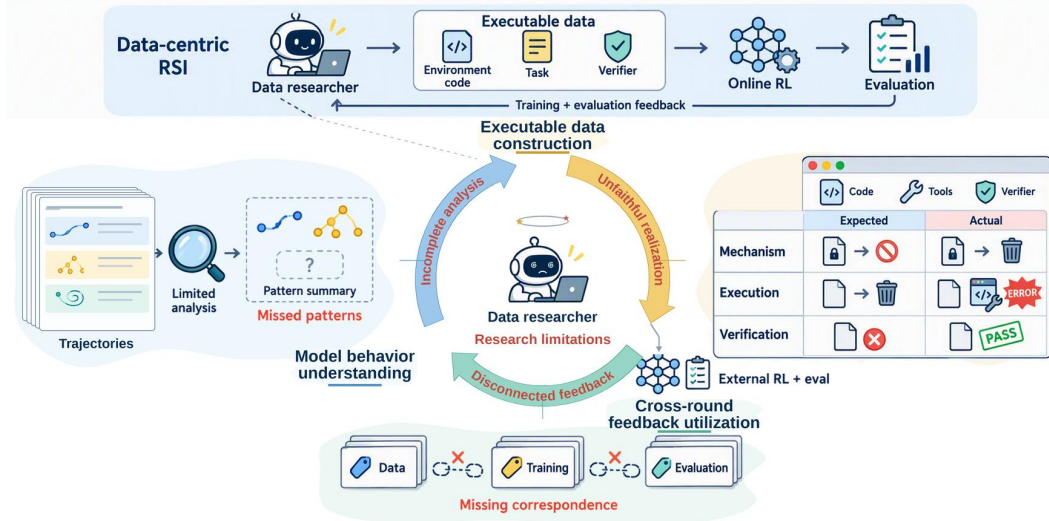


Figure 2: Three challenges in closing the data-centric RSI loop: incomplete behavioral analysis, unfaithful realization of executable training instances, and disconnected data–training–evaluation feedback. Illustrations and records are schematic.

provement: data, the execution harness, and model parameters. We focus on the interaction between the data and model surfaces, where an agent adapts training data as the learner evolves. Relatedly, RSIBench-Data (Meng et al., 2026) studies agents that revise training-data strategies across training runs and reveals a discovery–reliability gap: agents can discover useful strategies but struggle to improve them consistently. This raises a sharper question: when the learner itself changes, can feedback from one training round reliably guide data construction for the next?

We study this question in the context of training agents to use tools more effectively, as tool use is a fundamental capability for modern agents. Training such agents extend beyond generating static samples because training data take the form of executable instances that define state, tool behavior, tasks, and outcome verification. Systems for executable environment synthesis and scaling, including AgentScaler, EnvScaler, ScaleEnv, Agent World Model, and EnvFactory (Fang et al., 2025; Song et al., 2026; Tu et al., 2026; Wang et al., 2026; Xu et al., 2026), show that such resources can be constructed automatically. Building on this capability, we study data-centric recursive improvement in a stateful, multi-round online-RL setting, where a target model is iteratively updated using new data produced by a data-researcher agent. Specifically, starting from an empty pool of executable environments, the agent operates in a four-stage loop: analyzing the current target model’s behavior, constructing executable training instances, submitting them to a fixed external service for online-RL training and evaluation, and using the resulting training records and evaluation feedback to guide the next round. With each round continuing from the updated model, the training data and target policy evolve together over time. Reliable recursive improvement is therefore not a one-shot data-generation problem: it requires sustaining a coherent data loop as both continue to evolve. Our experiments show that even a modern agent with strong reasoning and coding capabilities (Codex with GPT-6 Astra) struggles to sustain improvement across rounds. The target model’s performance often plateaus or regresses, and some runs even end in training collapse.

Based on our experiments and observations, we identify three key stages in closing the data improvement loop and their associated failure modes:

- **Model behavior understanding.** *Incomplete analysis* of available traces yields a partial or biased understanding of the target model.
- **Cross-round feedback utilization.** *Disconnected feedback* prevents the agent from relating constructed data to realized training experience and subsequent target behavior.
- **Executable data construction.** *Unfaithful realization* produces environments that are logically inconsistent or misaligned with the targeted domain or capability.

To these ends, we introduce ENVOPT, a framework that addresses these failure modes through three corresponding components. To see patterns in target model behavior, a **traceable behavioral map** consolidates evidence across trajectories while preserving links to the source cases. To close the feedback loop across rounds, **target-conditioned context refinement** organizes relevant data, observed training dynamics, and subsequent evaluation outcomes around each training target. To faithfully realize each training target as executable data, **iterative environment construction** uses a planning, implementation, and refinement loop to bridge the last mile from target specification to logically consistent and behaviorally aligned environments. Extensive experiments across four agentic tool-use benchmarks demonstrate the effectiveness of ENVOPT for multi-round improvement. On Qwen3-4B, ENVOPT achieves substantial performance gains of +7.44, +16.67, +11.11, and +13.29 percentage points on τ^2 -Bench, BFCL-v4, ACEBench-Agent, and AppWorld, respectively. Across successive iterations, target model performance with ENVOPT follows a clear overall upward trend, whereas the same Codex researcher without these mechanisms produces irregular fluctuations with no sustained direction of improvement. Further analyses show that the three proposed designs improve their corresponding stages of the loop by increasing coverage of task requirements and failure mechanisms, improving the quality of realized RL signals, and producing executable environments that are more reliable and better aligned with their training targets.

2 RELATED WORK

Data-centric RSI. RSIBench-Data studies agents that revise training-data strategies for a separate target model under fixed training, evaluation, and resource controls (Meng et al., 2026). Autodata develops an agentic data scientist that iterates between data creation and quality assessment, including meta-optimization of the data scientist itself (Kulikov et al., 2026). DataMaster combines data discovery and transformation with tree search and cumulative memory while holding the learning algorithm fixed (Du et al., 2026). ENVOPT studies this research role for executable training environments: a researcher starts with an empty environment library and uses behavioral evidence to guide successive rounds of environment revision and online RL. Its focus is the evidence supporting those research decisions.

Executable environment synthesis for tool-calling agents. AgentScaler, EnvScaler, and ScaleEnv automate the construction of executable tools, state, and verifiable tasks (Fang et al., 2025; Song et al., 2026; Tu et al., 2026). Agent World Model scales database-backed synthetic environments for RL, while EnvFactory combines executable environment construction with topology-aware task synthesis (Wang et al., 2026; Xu et al., 2026). FACET grounds task instructions, solutions, and verifiers in shared executable state to preserve source intent and cross-artifact consistency (Shi et al., 2026). The ACE framework and WRIT further examine data accuracy, learner-relative complexity, diversity, and the evidence required for individual tool-use decisions (Zeng et al., 2026; Gu et al., 2026). These works establish construction and validation principles that ENVOPT builds on; our emphasis is using source-traceable behavioral analysis and actual-training feedback to decide which mechanisms to construct or revise.

Iterative RL and environment adaptation. PAIRED and Prioritized Level Replay adapt environment generation or selection using regret and learning-potential signals (Dennis et al., 2020; Jiang et al., 2021). For tool-use agents, RODS uses online reward statistics to synthesize structurally aligned tasks and update a replay pool (Fang et al., 2026). EnvHarness adapts existing environments from policy trajectories while retaining their original verifiers (Huang et al., 2026). SPADE trains a shared environment-designer and solver model through self-play, using hint-based regret, corpus grounding, and environment memory (Liu et al., 2026). Agent-World and AgentOmnia also use behavioral diagnosis to guide targeted task synthesis and self-evolution (Dong et al., 2026; Jiang et al., 2026). ENVOPT uses an external researcher to revise independent environments without the target backend or verifier. It connects the source evidence behind a diagnosis to realized program versions, training tasks, RL observations, and fixed-target evaluation, supporting inspection of what each intervention actually delivered.

3 AGENTS AS DATA RESEARCHERS: IDENTIFYING CHALLENGES IN DATA-CENTRIC RSI

We instantiate data-centric RSI as an iterative environment-generation and reinforcement-learning loop. An agent serves as the data researcher, constructing executable environments and tasks in which a target model trains with online RL. Feedback from training and evaluation informs the next round. We first formalize this process, then examine challenges in its research context and generated training instances.

Formalizing the data-research loop. A *research context* is the information available to an agent for research decisions, encompassing the research history, current state, and other decision-relevant information. Let C_t denote this context at round t and `DataResearcher` the agent that uses it to construct training instances. Each instance is a triple (e, x, v) , consisting of an executable environment e , a task x specifying a goal and initial conditions, and a verifier v assessing task completion. The agent’s output is a set D_t of n_t such instances:

$$D_t = \{(e_j, x_j, v_j)\}_{j=1}^{n_t} = \text{DataResearcher}(C_t). \quad (1)$$

The set D_t is then submitted for training and evaluation. Let π_t denote the target model at the start of round t and `TrainEval` the fixed online-RL and public-evaluation procedure. It returns the updated model π_{t+1} and feedback F_t comprising training records and evaluation results:

$$(\pi_{t+1}, F_t) = \text{TrainEval}(\pi_t, D_t). \quad (2)$$

Finally, `UpdateContext` integrates the submitted set and its feedback with the prior research context to guide the next round:

$$C_{t+1} = \text{UpdateContext}(C_t, D_t, F_t). \quad (3)$$

In this setting, the loop starts from an empty environment library. Guided by feedback and the evolving research context, the data researcher repeatedly creates new environments or revises existing ones, with the goal of improving the target model over successive training rounds.

3.1 INCOMPLETE BEHAVIORAL ANALYSIS

The research context must preserve task requirements and model behavior to support training-target selection. To examine what survives this analysis, we give an analyst agent (Astra/high) complete access to the initial public-eval trajectories of Qwen3-8B reasoning: 1,200 BFCL, 417 τ^2 , and 75 ACEBench-Agent trajectories. The instruction requests an analysis and specifies report fields, while leaving the reading strategy to the agent. We measure report coverage of source-grounded task requirements and failure mechanisms, crediting full preservation of their consequential conditions and dependencies (Section 5.3).

The analysis reveals a substantial gap: task-requirement coverage is 61.5–82.5%, while failure-mechanism coverage ranges from only 39.4% on BFCL to 67.5% on ACE. These automatic, task-averaged scores use the same final paired evaluation as Section 5.3; the remainder includes both partial descriptions and wholly missing mechanisms. Full-corpus access thus leaves substantial behavioral information incompletely represented. Coverage is lower on the larger corpora in this comparison, although benchmark composition and corpus size vary together, preventing a causal size claim.

The research context therefore needs to preserve the mechanisms and conditions that matter for the next data decision, with evidence available for inspection.

3.2 DISCONNECTED DATA, TRAINING, AND EVALUATION

Updating the research context requires connecting intended training targets, tasks actually trained, and subsequent evaluation outcomes. In ACE R9, one revision contributed zero of its ten planned tasks. We further audited 28 iterations of the original 4B studies, measuring the fraction of trained tasks with nonconstant rewards across eight valid rollouts—the within-task contrast used by GRPO. Among complete, valid groups, the task-weighted rates were 74.6%, 42.2%, and 37.2% on τ^2 ,

BFCL, and ACE. These checks distinguish whether an intervention reached training and whether its rollouts supplied reward contrast, motivating explicit linkage to its intended target and evaluation outcomes. Section 5.4 compares the per-round signal with Codex-Vanilla.

3.3 INCOMPLETE ENVIRONMENT IMPLEMENTATION

A retail-exchange environment required matching identical items to different replacements, yet all six inspected tasks assigned the same replacement to both items. Checking the generator against the specification confirmed that the required branch was absent. This exposes a gap in constructing the output triples: a runnable instance need not realize the intended training mechanism. Auditing environment behavior, task requirements, and verifier scores tests whether the three components jointly deliver the training target. [TODO: Cross-benchmark audit: 20 environments each on τ^2 , BFCL, and ACE; eight rollouts per environment; LLM judgments of target implementation, interaction, and verifier defects. Insert observed defect rates.]

4 METHOD

ENVOPT instantiates the data-research loop in Equations 1–3 (Figure 3). Its two core designs organize the research context C_t : a *traceable behavioral map* preserves task requirements and failure mechanisms with links to source trajectories; *target-conditioned context refinement* uses target-linked feedback to build decision-ready context for comparing intended training, actual RL experience, and target-evaluation outcomes. Guided by this context, specification-driven construction and independent review produce the environment–task–verifier set D_t . The training-and-evaluation service remains fixed.

The following describes the maintained framework and labels a standalone analysis refinement. Section 5 distinguishes these from the earlier protocols underlying the reported training curves.

4.1 TRACEABLE BEHAVIORAL MAP

The map separates a persistent evidence layer from its compact reading view. Let $T_{t,i}$ contain the complete public-eval episodes for task i , including instructions, tool definitions, and observed interactions. Extraction and source review produce an immutable mechanism ledger:

$$\mathcal{L}_t = \bigcup_i \text{ExtractReview}(T_{t,i}). \quad (4)$$

Each record links an environment mechanism, task requirement, or observed failure/recovery to its source and applicable conditions.

Bounded hierarchical merging builds a draft by factoring shared premises while retaining distinct actors, scopes, dependencies, and recovery branches:

$$R_t^{(0)} = \text{FactorMerge}(\mathcal{L}_t). \quad (5)$$

The draft is checked against the retained evidence, not only earlier summaries. Repair locally corrects conflicting claims; Residual extracts and consolidates only relations still missing from the corrected report:

$$\begin{aligned} \bar{R}_t &= \text{Repair}(R_t^{(0)}, \mathcal{L}_t), \\ R_t &= \bar{R}_t \oplus \text{Residual}(\mathcal{L}_t, \bar{R}_t). \end{aligned} \quad (6)$$

Here $\oplus$ appends the missing distinctions without rewriting preserved content; an already-covered record must instead be matched to an existing passage. These are model-based checks, not a semantic-completeness guarantee.

The report retains the four fields *domain*, *environment*, *task behavior*, *failure behavior*. It enters the research context C_t with links back to original trajectories. This coverage-checked variant is evaluated offline in Section 5.3, separately from the earlier analysis used in the historical training runs.

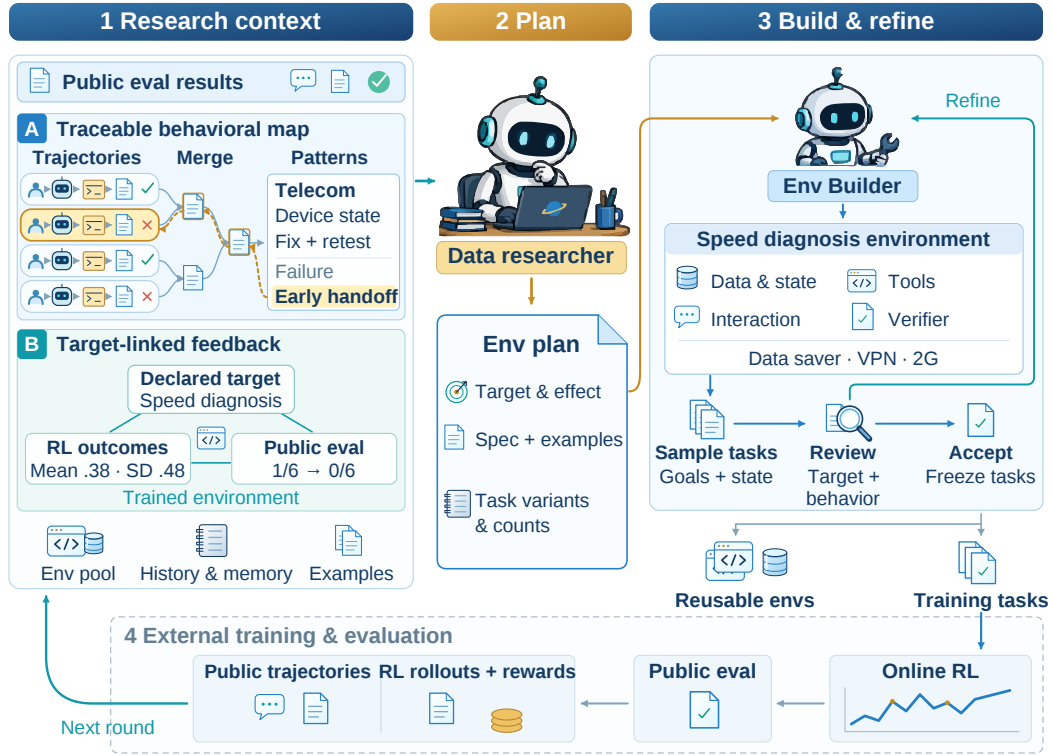


Figure 3: ENVOPT’s research loop. A highlighted pattern traces back through the merge tree; linked target, training, and public-eval evidence guide the data researcher. Sampled tasks are independently reviewed and frozen before RL. Curves are schematic; B shows one historical example.

4.2 TARGET-CONDITIONED CONTEXT REFINEMENT

Access to evidence does not ensure its use in research decisions. We target this *information-utilization gap* with *decision-ready context*: evidence organized for joint interpretation around a shared behavioral target. Our *target-linked feedback* provides the context-level intervention, explicitly connecting what was intended, what was trained, and how target behavior changed.

For each intervention, the researcher declares a target requirement and expected effect, anchored to public-eval tasks. Versioned triples in D_t and their task IDs link this target to actual RL groups and rollouts, forming a target-conditioned evidence view:

$$\underbrace{\text{Training target}}_{\text{requirement, expected effect}} \longleftrightarrow \underbrace{\text{Actual training}}_{\text{versioned triples, RL rollouts}} \longleftrightarrow \underbrace{\text{Target evaluation}}_{\text{linked public tasks}}. \quad (7)$$

The view distinguishes planned, accepted, and actually trained data, pairing training-time reward statistics and within-task reward contrast with linked public-eval outcomes and trajectories. The researcher compares these observations with the expected effect, recording agreements, mismatches, and unresolved questions in C_{t+1} . This assessment guides revisions to mechanisms, challenge, or sampling, and decisions to retain or pause environments. The design guides evidence use rather than adding otherwise inaccessible feedback; the links do not establish causal effects of individual edits.

4.3 ITERATIVE ENVIRONMENT CONSTRUCTION

We construct training environments through an iterative cycle of planning, implementation, and review, revising either the implementation or the plan as needed. The researcher selects training targets and task allocations, then provides specifications with success, failure, and recovery cases. Persistent Env Builders receive these specifications rather than raw public-eval or RL corpora.

Each builder implements candidate (e, x, v) triples, with task requests and verifiers sharing the same goal data. Execution checks and independent review examine the specification, full program, and all tasks in the batch. They assess whether the intended decisions and variations are implemented and whether information access, tool effects, and verifier rewards are consistent. Implementation findings return to the same builder for revision and renewed review, retaining prior findings; specification ambiguities return to the researcher to revise the plan.

Accepted triples are frozen and retained while the researcher replans any remaining allocations, subject to construction budgets. The accepted set D_t enters training once its admission requirement is met. This local construction cycle precedes the outer training and feedback loop; review remains fallible, not a proof of program correctness.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Evaluation and training. We evaluate on τ^2 -Bench (Barres et al., 2026), BFCL-v4 (multiturn) (Patil et al., 2025), ACEBench-Agent (Chen et al., 2025), and AppWorld (Trivedi et al., 2024). The first three benchmarks use 1:1 public/private splits; AppWorld uses its train split for public evaluation and test-normal split for private evaluation. Public evaluation informs environment evolution and checkpoint selection, while private evaluation is withheld from the evolution process and used for final performance comparisons. Target models train with GRPO (Shao et al., 2024).

Implementation and baselines. Our target models are Qwen3-4B-Instruct-2507 and Qwen3-8B (Yang et al., 2025), with reasoning enabled for the latter. Our data researcher agent, DataResearcher, is instantiated with Codex powered by GPT-6 Astra. We grant the agent access to the GPT-5.5 API and 4 A100 GPUs during evolution. We compare against three baselines: vanilla Codex as an autonomous data researcher (Codex-Vanilla); Codex with an example code repository provided (Codex-E); and training on a fixed environment set generated by DataResearcher without revision (Fixed-Env). All researcher agents in these baselines are powered by GPT-6 Astra.

5.2 MAIN RESULTS

Strong coding agents struggle with continual data evolution. Despite pairing GPT-6 Astra with the Codex harness, Codex-Vanilla shows early gains followed by saturation or regression (Figure 1). On BFCL, public performance peaks at 30.00% in R3 and falls to 25.42% by R10. On τ^2 , it finishes at 21.58%, below its 27.10% starting score.

A plausible bottleneck is maintaining the learning value of generated environments as the target model changes. This requires updating training targets from behavioral feedback and faithfully implementing the corresponding mechanisms, the challenges identified in Section 3. Incomplete implementations or poorly updated targets may instead produce executable environments that offer little additional learning signal.

Substantial gains across four benchmarks. ENVOPT improves 4B public-evaluation performance by +8.63, +22.58, +30.00, and +26.75 percentage points over the base model on τ^2 -Bench, BFCL-v4 (multiturn), ACEBench-Agent, and AppWorld, respectively. These results use the earliest public maximum within the displayed rounds. The contrast is clearest on BFCL: ENVOPT reaches 45.00% in R9, well beyond Codex-Vanilla’s 30.00% peak in R3. On both benchmarks with baseline curves, ENVOPT maintains higher scores throughout the shared R3–R9 rounds. All four displayed endpoints remain above their starting scores. Table 1 reports test performance at the validation-selected checkpoints.

5.3 BEHAVIORAL COVERAGE AND REPORT COMPACTNESS

We now test whether a structured analysis process reduces the information loss identified in Section 3.1. On the same 1,692 frozen initial Qwen3-8B public-eval trajectories, we compare the fixed coding-agent reports (Astra/high) with a standalone, coverage-checked refinement of the behavioral map (GPT-5.5/high). The refinement analyzes complete task evidence, consolidates shared mech-

Table 1: Test performance (%) at checkpoints selected on the public validation set. Green values show Δ from None (pp); Avg. averages all four benchmarks. Indented rows enable only the named component: BM, traceable behavioral map; CR, target-conditioned context refinement; EC, iterative environment construction. Dashes indicate unreported results.

Method	τ^2 -Bench	BFCL-v4	ACEBench-Agent	AppWorld	Avg.
<i>Qwen3-4B-Instruct-2507</i>					
None	31.89	25.42	41.11	8.93	26.84
Codex-Vanilla	—	—	—	—	—
Codex-E	—	—	—	—	—
Fixed-Env	—	—	—	—	—
ENVOPT	39.33 +7.43	42.08 +16.67	52.22 +11.11	22.22 +13.29	38.96 +12.13
↪ BM-only	—	—	—	—	—
↪ CR-only	—	—	—	—	—
↪ EC-only	—	—	—	—	—
<i>Qwen3-8B (reasoning enabled)</i>					
None	28.54	44.42	43.89	—	—
Codex-Vanilla	—	—	—	—	—
Codex-E	—	—	—	—	—
Fixed-Env	—	—	—	—	—
ENVOPT	—	—	—	—	—
↪ BM-only	—	—	—	—	—
↪ CR-only	—	—	—	—	—
↪ EC-only	—	—	—	—	—

anisms, and checks the compact report for omitted conditions. It receives neither the reference labels nor the coding agent’s report prose. We extend the same protocol to 243 initial public-eval trajectories of Qwen3-4B-Instruct-2507 on AppWorld.

Measure retained meaning, not compression alone. Frozen, source-grounded reference units describe task requirements and failure mechanisms. An LLM judge scores anonymized report pairs as full, partial, missing, or contradicted. Only full preservation of the mechanism and its consequential scope or dependency receives credit. We average units within each task and then tasks within each corpus; word count separately measures compactness. References are generated before seeing candidate reports, not inferred from either method’s preferred categories.

Higher coverage in shorter reports. The refined map reaches 88.1–99.5% task-requirement coverage and 71.3–88.4% failure-mechanism coverage, while using 12–29% fewer words (Figure 4a). It exceeds the paired grades of the coding-agent reports on all four corpora; the ordering also holds under subset–family-balanced aggregation. A concrete ACE example shows why this is more than naming more error types: an inbox-cleanup timeout label omits the user-authorization–deletion–retry dependency, which the refined report preserves. Both graders support partial versus full coverage for that example. Not every case improves.

These development-set results support more comprehensive, compact behavioral analysis in this setting. They do not isolate the hierarchy from the added coverage checks, match backbones or computation, or establish downstream RL gains. The diagnostic is separate from the historical 4B training runs.

5.4 TRAINING SIGNAL ACROSS EVOLUTION

We audit the saved RL reward groups from the original 4B experiments and Codex-Vanilla. A task provides within-group reward signal when its eight valid rollouts receive nonconstant rewards; we report the fraction of such tasks among complete, valid groups in each round (Figure 4b). This measures the relative reward contrast used by GRPO, retaining partial-credit rewards and excluding incomplete groups rather than treating them as failures.

Training success can exhaust the learning signal. On τ^2 , every Codex-Vanilla task in R9 and R10 receives eight rewards of one, leaving no within-group reward contrast. On BFCL, only 2

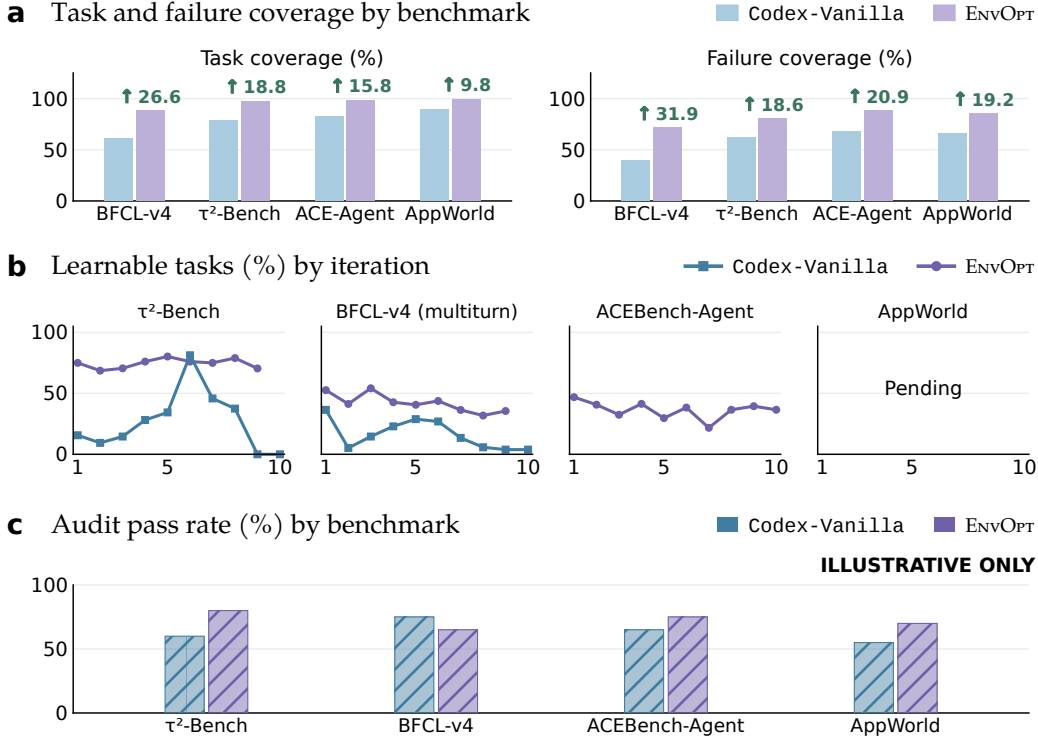


Figure 4: Research-process diagnostics. (a) Task and failure coverage of the coding-agent reports and refined map; $\uparrow$ labels show gains over Codex-Vanilla in percentage points. (b) Learnable tasks (nonconstant rewards across eight valid rollouts) by iteration; AppWorld remains pending in this panel. (c) Illustrative environment-audit pass rates, using placeholder data rather than experimental results.

of 52 tasks retain reward variation in each of these rounds. In comparison, ENVOPT retains signal on 70.5% of valid τ^2 tasks and 35.5% of BFCL tasks in its final recorded round, R9. These observations motivate linking generated tasks to their actual training outcomes so the researcher can detect reward saturation and refresh its targets. They do not isolate the causal effect of context refinement.

5.5 ENVIRONMENT IMPLEMENTATION QUALITY

[TODO: Audit 20 sampled environments per benchmark, with eight rollouts each, using an independent LLM judge to flag implementation, interaction, and verifier defects. Replace the illustrative values in Figure 4c with measured environment-level audit pass rates (no defects flagged in the inspected rollouts).]

5.6 REMAINING CONTROLS AND UNCERTAINTY

Controls still required. The available baseline records cover Codex-Vanilla; Codex-E and Fixed-Env comparisons are not yet reported. The offline diagnostic measures report quality, not a budget-matched isolation of the two designs. Such controls should match researcher, information, and training budgets. For behavioral understanding, compare selective case reading, a flat summary, and the source-traceable hierarchy, counting both analysis and researcher tokens. For context refinement, compare the same underlying records with and without target-linked organization, measuring evidence use and subsequent decisions under matched budgets and construction workflows. The proposed attention-guidance effect has not been isolated. An end-to-end native coding-agent control should retain equivalent tools and service access; the separately reported Codex curves do not establish such a matched control. Secondary controls compare a fixed library with fresh tasks, sampler/allocation-only adaptation, and program revision, or remove maintained experience. These

causal controls remain unmeasured. SPADE-style or EnvHarness-style controls must additionally disclose differences in target-code and public-eval-feedback access.

Uncertainty and cost. Paired-task uncertainty is estimated using 10,000 bootstrap resamples of private-eval tasks within fixed subsets, keeping each task’s three repeats together and preserving the benchmark aggregation. These intervals measure finite-private-eval-set uncertainty conditional on a trained run, not training-seed variability. The recorded rounds contain 846, 809, and 626 distinct training Tasks for τ^2 , BFCL, and ACE respectively, with eight episodes per Task. Round counts do not fix training budgets: native ACE uses 96 Tasks in R1, 38 in R9, and 52 in R10. Final cost comparisons must also count analysis, planning, failed implementations, screening, serving, and evaluation, including queues and restarts. We make no end-to-end efficiency claim from task counts alone.

6 CONCLUSION

We study agents as data researchers that iteratively construct executable environments for model training. ENVOPT organizes this process around a traceable behavioral map and target-conditioned context refinement, supported by iterative environment construction. Our analyses identify gaps in behavioral understanding, feedback linkage, and environment implementation. Preliminary training results show useful checkpoints but also regressions, motivating controlled studies of sustained improvement.

AI USE STATEMENT

Generative AI assisted source-code and literature analysis, methodological formulation, drafting, and figure preparation. The research system uses language models for trajectory analysis, planning, program synthesis, solvability review and user simulation; generated-task rewards in the studied cohort are programmatic. Training results use existing experiment records. The offline diagnostic additionally generated and graded analysis reports from frozen public-evaluation trajectories; it did not rerun benchmark tasks or training. [TODO: The authors must verify this statement against the complete research history and review all AI-assisted claims and artifacts before submission.]

REPRODUCIBILITY STATEMENT

The research loop is specified in Section 3, construction interfaces and admission rules in Section 4, and evaluation access in Section 5. [TODO: Add the frozen implementation revision, resolved run configurations, complete artifacts, and an anonymous code supplement after the experiments are finalized.]

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